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ROLL NO - CSB24021

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# Address class
class Address:
    def __init__(self, street, city, zipCode):
        self.street = street
        self.city = city
        self.zipCode = zipCode

    def display(self):
        return f"{self.street}, {self.city} - {self.zipCode}"

# Student class (Composition: HAS-A Address)
class Student:
    def __init__(self, name, age, address):
        self.name = name
        self._age = None # protected attribute
        self.age = age # setter will validate
        self.address = address
        self.courses = [] # mutable list

    # Property for age
    @property
    def age(self):
        return self._age

    @age.setter
    def age(self, value):
        if value <= 0 or value > 120:
            raise ValueError("Age must be between 1 and 120")
        self._age = value

    # Method to add course
    def add_course(self, course):
        self.courses.append(course)

    # Display method
    def display(self):
        print("---- Student Details ----")
        print(f"Name: {self.name}")
        print(f"Age: {self.age}")
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        print(f"Address: {self.address.display()}")
        print(f"Courses: {' , '.join(self.courses) if self.courses else
'No courses'}")

# Inheritance + Overriding
class ScholarshipStudent(Student):
    def __init__(self, name, age, address, scholarshipAmount):
        super().__init__(name, age, address)
        self.scholarshipAmount = scholarshipAmount

    # Override display()
    def display(self):
        print("---- Scholarship Student Details ----")
        super().display() # call parent method
        print(f"Scholarship Amount: {self.scholarshipAmount}")

# ----- Testing -----

# Create Address object
addr1 = Address("ABC Road", "Guwahati", "781001")

# Create Student object
s1 = Student("Rahul", 20, addr1)
s1.add_course("Python")
s1.add_course("Data Structures")

# Create ScholarshipStudent object
s2 = ScholarshipStudent("Priya", 22, addr1, 50000)
s2.add_course("Machine Learning")

# Display details
s1.display()
print()
s2.display()

```

OUTPUT -

```
OPEN EDITORS
  collect.py
  student_system.py
  library.py
PYTHON
  collect.py
  first.py
  library.py
  product.py
  student_system.py
  time complexity.c
  time complexity.exe

[Running] python -u "c:\Users\prerana\OneDrive\Desktop\PYTHON\
---- Student Details ----
Name: Rahul
Age: 20
Address: ABC Road, Guwahati - 781001
Courses: Python, Data Structures

---- Scholarship Student Details ----
---- Student Details ----
Name: Priya
Age: 22
Address: ABC Road, Guwahati - 781001
Courses: Machine Learning
Scholarship Amount: 50000

[Done] exited with code=0 in 0.073 seconds
```